MCO Wave Parameter Library

ADDENDUM REV A — Variable Sweep Cruise Modes (Tweaks 3 & 4)

MCO-1A Piloted SAR Ornithopter · Issue date: 2026-06-02

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1. Purpose of This Addendum

This addendum adds two new mechanisms to the MCO wave parameter library, derived from analysis of partial-stow cruise performance and the identified path to 120-knot GA cruise speeds without auxiliary propulsion. It supplements the existing locked Rev G wave library and does not alter any previously confirmed modes (metachronal cascade, stoop sequence, or control architecture).

Tweak 3 introduces **continuous variable sweep as the primary speed control**. Tweak 4 introduces **dihedral bias at high sweep angles** to enable efficient thrust generation in the swept-wing sprint regime. Both are implemented entirely in FCC software and wave library parameters — no hardware changes are required.

2. Physics Background

2.1 Why variable sweep raises cruise speed

The partial-stow analysis established that stowing W1+W2 at 200° raises the best-range speed from 32 kt (full wings) to 46 kt (W3+W4 only). The mechanism is straightforward: smaller active wing area requires higher dynamic pressure — i.e. higher speed — to generate the same lift. The aircraft's minimum-drag operating point shifts up the speed axis.

Extending this to all eight wings swept to 135° reduces the effective projected area from 30.7 m² to approximately 21.7 m². At 120 kt (61.7 m/s) the required CL is only 0.058 — well within the delta-wing LEV regime available at 135° sweep. Total drag at 120 kt is 1,092 N and required power is 67 kW.

2.2 Why flapping at 135° sweep still generates thrust

At low sweep angles the flap stroke oscillates the wing vertically through a wide arc, producing lift on the downstroke and thrust via the leading-edge suction mechanism. At 135° sweep the geometry changes: the wing now points aft-outboard, and a vertical oscillation produces a force vector that is mostly drag and only partly thrust.

Tweak 4 corrects this by adding a negative pitch offset (dihedral bias) when sweep exceeds 120°. This tilts the effective flap stroke plane so the wing oscillates in a direction that creates a useful forward-thrust component. The mechanism is directly analogous to the wing motion of swifts (Apus apus) in high-speed cruise: small-amplitude pitch variation of a highly-swept narrow wing, rather than a wide arc flap. At high dynamic pressure, even small angular motions produce substantial thrust because force scales with q × S × CT.

Key numbers (MCO-1A, 300 kg, 135° sweep, all 8 wings):
Effective lift area: 21.7 m² · CL at 120 kt: 0.058 · Total drag at 120 kt: 1,092 N · Power: 67 kW
Thrust available (CT=0.15 conservative): 4,561 N · Margin: +3,469 N · Result: 120 kt achievable
CT uncertainty range: 0.08 (worst) → 0.25 (optimistic). MCO-1D flight test resolves this before MCO-1A build.

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3. Tweak 3 — Continuous Variable Sweep Speed Control

3.1 Concept

The existing wave library treats wing sweep as a mode-specific parameter: wings are set to a fixed sweep angle appropriate for the active flight mode (60° for normal flight, 200° for stoop). Tweak 3 makes sweep angle a continuously variable function of the pilot's speed lever position, exactly as variable-sweep aircraft (F-14 Tomcat, Su-24) use wing sweep as an aerodynamic speed control.

The FCC reads the speed lever position and the pitot airspeed and computes a target sweep angle for each wing pair. Wings sweep progressively aft as the speed demand increases. Transition is smooth — no discrete mode switch. The stall-protection floor remains active throughout: if actual airspeed falls below V_s+10% for the current sweep configuration, the FCC automatically begins deploying wings forward.

3.2 Sweep schedule

Speed lever	Airspeed band	W1 sweep	W2 sweep	W3 sweep	W4 sweep	Mode label
Detent (glide)	<16 kt	60°	60°	60°	60°	Full deploy — soar
0–30%	16–32 kt	60°	60°	60°	60°	Slow cruise — heron
30–50%	32–46 kt	60°→120°	60°	60°	60°	W1 sweeping aft
50–65%	46–68 kt	200°*	200°*	60°	60°	W3+W4 cruise — swift
65–80%	68–97 kt	200°*	200°*	60°→135°	60°→135°	All sweeping to 135°
80–95%	97–120 kt	200°*	200°*	135°	135°	GA sprint — 135° delta
95–100%	>120 kt	200°*	200°*	165°→200°*	135°→165°	Pre-stoop sweep
STOOP switch	—	200°*	200°*	200°*	200°*	Full stow — stoop

* 200° with spring-loaded root fairing closed (Tweak 1, separate addendum). Fairings are required for 68+ kt operation.

3.3 FCC implementation

The sweep schedule is implemented as a lookup table in the wave library Python module on the Raspberry Pi Zero 2W. The table maps (speed_lever_position, pitot_airspeed) → (W1_sweep_target, W2_sweep_target, W3_sweep_target, W4_sweep_target). The FCC sends sweep targets to each ODrive over CAN-FD. Sweep rate is limited to 30°/s during normal speed transitions (60°/s is reserved for emergency stoop stow).

Parameter	Value	Notes
Sweep rate — normal	30°/s max	Smooth speed transitions
Sweep rate — stoop stow	60°/s min	Emergency, unchanged from base spec
Sweep deadband	±2°	Prevents hunting at steady state
Stall protection floor	Vs + 10%	Auto-deploys wings if airspeed drops
Lever hysteresis	±5%	Prevents oscillation at threshold
Transition smoothing	2nd-order filter, τ=0.8s	Prevents abrupt pitch trim change

3.4 Pitch trim compensation

Sweeping wings aft shifts the aerodynamic centre rearward and reduces total lift, causing a nose-down pitching moment. The FCC automatically compensates by commanding a positive pitch increment on all remaining deployed wings as sweep increases. The compensation schedule is:

Sweep angle	Pitch trim offset	Rationale
60°–90°	0° (baseline)	Normal flight trim
90°–120°	+2°	Slight nose-up to compensate CP shift
120°–135°	+4°	Moderate CP rearward shift
135°–165°	+6°	Significant CP rearward — full compensation
165°–200°	Stoop pitch DoF	Active teardrop — see STOOP-2A spec

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4. Tweak 4 — Dihedral Bias at High Sweep Angles

4.1 Concept

At 135° sweep the wing points aft-outboard at 45° to the fuselage axis. A conventional flap stroke (vertical oscillation about the root pivot) produces a force vector that is largely perpendicular to the wing span — which at 135° sweep is mostly sideways and slightly forward, not usefully rearward as required for thrust.

Tweak 4 adds a negative pitch offset (dihedral bias) to all wings when sweep exceeds 120°. This rotates the wing's chord line downward relative to the fuselage datum. The resulting flap stroke now oscillates in a plane tilted forward relative to vertical, producing a force vector with a larger forward component. The mechanism is identical to how swifts generate thrust in high-speed cruise: rapid small-amplitude pitch variations of a highly-swept narrow wing rather than wide arc flapping.

The dihedral bias is applied via the existing pitch servo on each wing root. No hardware change is required — the pitch servo range (–5° to +50°) already covers the required offset. The bias is a static offset added to the instantaneous pitch command from the wave library.

4.2 Dihedral bias schedule

Sweep angle	Dihedral bias	Effective flap plane	Thrust component
60°–120°	0° (none)	Vertical — normal	Standard ornithopter
120°–130°	−4°	4° forward tilt	Increasing forward thrust
130°–140°	−8°	8° forward tilt	Good thrust efficiency
140°–150°	−12°	12° forward tilt	Optimal for 100–120 kt
150°–165°	−15°	15° forward tilt	High-speed pre-stoop
165°–200°	Stoop pitch DoF active	Active control	See STOOP-2A

4.3 Interaction with flap amplitude

At high sweep + dihedral bias the optimal flap amplitude is reduced. The wave library scales flap amplitude as a function of sweep angle:

Sweep	Flap amplitude	Frequency (MCO-1A)	Notes
60°	±80° (full)	0.11–0.40 Hz	Normal ornithopter flight
90°	±60°	0.11–0.35 Hz	Slight reduction for efficiency
120°	±40°	0.15–0.50 Hz	Amplitude reducing, freq rising
135°	±20°	0.20–0.80 Hz	Swift-mode oscillation
150°	±10°	0.30–1.20 Hz	Small rapid pitch variation
165°	±5°	0.50–2.00 Hz	Minimal motion, high q

Note: MCO-1D frequencies are 7.5× higher than MCO-1A values shown above. The MCO-1D drone validates CT in this regime before MCO-1A build. The drone flies the same 135° sweep + dihedral bias modes at proportionally higher frequency, and the measured CT transfers directly.

4.4 FCC implementation

Parameter	Value	Notes
Bias onset threshold	120° sweep	No bias below this angle
Bias rate limit	2°/s per degree of sweep change	Smooth transition
Bias source	Pitch servo offset command	Added to instantaneous pitch cmd
Override	Stall protection takes priority	Bias removed if Vs approached
Asymmetric use	Port/stbd can differ ±2°	Roll authority at high speed
MCO-1D validation	Required before MCO-1A build	Measure actual CT vs model

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5. Revised MCO-1A Speed Envelope

The following table supersedes prior performance summaries. Values marked (validated) follow from confirmed aerodynamic relationships. Values marked (model, requires CT validation) depend on the swept-oscillation thrust coefficient established by MCO-1D flight testing.

Mode	Wing config	Stall	Best range	Sprint	Status
Full-wing soar	W1–W4 at 60°	16 kt	32 kt	43 kt	Validated physics
W3+W4 cruise	W1+W2 faired at 200°	22 kt	46 kt	68 kt	Validated physics
135° delta sprint	All wings at 135°, dihedral bias active	31 kt	49 kt	97–120 kt	Model, CT validation req.
STOOP-2A	All faired at 200°, active pitch/sweep DoF	—	—	156 kt*	Drag calculation

* 156 kt stoop assumes enclosed teardrop fuselage (CD·S = 0.12 m², Pipistrel Virus SW class) with all wings faired flush. Compressibility effects not modelled — actual terminal velocity will be limited by structural and pilot tolerances before aerodynamic limit.

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6. MCO-1D Validation Requirements

Tweaks 3 and 4 introduce a thrust mechanism (swept small-amplitude oscillation) that has not previously been demonstrated on this airframe. The CT value determines whether 120 kt level flight is achievable (CT ≥ 0.20) or whether a shallow descent is needed (CT = 0.10–0.15 gives ~100 kt level, 120 kt in 5°). The MCO-1D resolves this before the MCO-1A is built.

Test	Procedure	Success criterion	Applies to MCO-1A as
T3-1: Partial stow cruise	Stow W1+W2, fly W3+W4 only at increasing speed lever	Sustains 68 kt level flight for 30 sec	Confirms W3+W4 cruise mode
T3-2: Progressive sweep	Advance speed lever through full range, note speed achieved	Speed increases smoothly with lever position	Validates sweep schedule
T4-1: Dihedral bias onset	At 30 kt, sweep all wings to 135°, apply –12° bias, hold	Airspeed increases without altitude loss	Confirms thrust from swept oscillation
T4-2: CT measurement	Level flight at 135° sweep + bias, log airspeed vs freq	Fit CT from T = CT·q·S at measured speed	Primary scaling input for MCO-1A 120 kt
T4-3: Stall check at 135°	Reduce speed lever slowly until stall at 135° config	Stall above 30 kt (model: 31 kt)	Validates delta-wing CLmax model

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7. Wave Library Parameter Additions

The following new parameters are added to the MCO wave library Python module. All existing parameters remain unchanged.

7.1 Sweep schedule table (Tweak 3)

```
# TWEAK 3 — Variable sweep speed control # Added to wave_library.py, MCO_FCC module SWEEP_SCHEDULE = [ # (lever_pct, V_kt_min, V_kt_max, W1_sw, W2_sw, W3_sw, W4_sw) ( 0, 0, 32, 60, 60, 60, 60), # Full deploy ( 30, 32, 46, 120, 60, 60, 60), # W1 sweeping ( 50, 46, 68, 200, 200, 60, 60), # W3+W4 cruise ( 65, 68, 97, 200, 200, 135, 135), # All sweep to 135 ( 80, 97, 120, 200, 200, 135, 135), # GA sprint ( 95, 120, 999, 200, 200, 165, 165), # Pre-stoop ] SWEEP_RATE_NORMAL = 30 # deg/s — smooth transitions SWEEP_RATE_STOOP = 60 # deg/s — emergency stow (unchanged) SWEEP_DEADBAND = 2 # deg — prevents hunting LEVER_HYSTERESIS = 5 # % — prevents threshold oscillation TRANSITION_TAU = 0.8 # s — 2nd-order filter time constant STALL_MARGIN_KT = 10 # % above Vs — auto-deploy trigger
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7.2 Pitch trim compensation table (Tweak 3)

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# TWEAK 3 — Pitch trim vs sweep angle PITCH_TRIM_SCHEDULE = [ # (sweep_deg, pitch_offset_deg) ( 60, 0.0), ( 90, 0.0), ( 120, +2.0), ( 135, +4.0), ( 150, +6.0), ( 165, +6.0), ] # interpolated linearly between breakpoints
```

7.3 Dihedral bias schedule (Tweak 4)

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# TWEAK 4 — Dihedral bias vs sweep angle DIHEDRAL_BIAS_SCHEDULE = [ # (sweep_deg, dihedral_offset_deg) ( 120, 0.0), # onset threshold ( 130, -4.0), ( 140, -8.0), ( 150, -12.0), ( 165, -15.0), ] # interpolated linearly; clamped at 165 deg DIHEDRAL_BIAS_RATE = 2.0 # deg/s per deg/s of sweep rate DIHEDRAL_ASYM_LIMIT = 2.0 # deg — max port/stbd asymmetry for roll
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7.4 Flap amplitude schedule (Tweak 4)

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# TWEAK 4 — Flap amplitude vs sweep angle FLAP_AMP_SCHEDULE = [ # (sweep_deg, amplitude_deg) ( 60, 80), # full arc ( 90, 60), ( 120, 40), ( 135, 20), # swift-mode ( 150, 10), ( 165, 5), ] # interpolated linearly between breakpoints
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8. Interaction with Existing Wave Library Modes

Existing mode	Interaction	Action required
Metachronal 90° cascade	Unchanged. Active when sweep < 120° and flap amplitude > 40°.	None.
Asymmetric sweep turns	Still active at all sweep angles. At 135° sweep, asymmetric sweep ±10° gives differential drag for yaw control.	Verify yaw authority adequate at 135° in T3-2.
STOOP-1 (progressive stow)	Unchanged. STOOP switch bypasses Tweak 3 schedule and commands 60°/s sweep to 200°.	None — STOOP switch has priority.

Existing mode	Interaction	Action required
STOOP-2A (active teardrop)	Unchanged. Dihedral bias is removed when stoop mode is active (pitch DoF becomes free for control).	Add: clear dihedral bias on STOOP-2A entry.
STOOP-4A (M-shape pullout)	Unchanged. W3+W4 deploy from 200° to 90° for LEV pullout.	None.
Glide detent	At glide detent, wings lock at 60° (max area, best glide AoA). Sweep schedule suspended.	None.

9. Open Items and Pending Work

#	Item	Dependency	Priority
O-1	MCO-1D flight test of swept oscillation modes (T4-1 through T4-3) to measure CT	MCO-1D build complete	CRITICAL — gates MCO-1A 120 kt claim
O-2	Tune TRANSITION_TAU and LEVER_HYSTERESIS values in hardware-in-loop simulation	FCC software build	High
O-3	Validate pitch trim compensation schedule on MCO-1D	T3-2 flight test	High
O-4	Structural review of MCO-1A wing root bearing under combined sweep+dihedral loads	MCO-1A detailed design	Medium
O-5	Assess asymmetric dihedral bias as roll control at 135° sweep	T4-2 flight test data	Medium
O-6	Spring-loaded root fairings (Tweak 1) design and mass confirmation	MCO-1A fuselage design	Medium